NPCs and Player Flow — Applied Thesis

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The use of non-player characters (NPCs) has been used since classic tabletop games. The principle is to create a character that can act as a human player's substitute in the game world. NPCs can serve a variety of purposes, such as providing information, offering quests or missions, or serving as adversaries for players to battle against. As described by psychologist Mihaly Csikszentmihalyi, flow is a state of complete engagement and immersion where individuals are fully absorbed in an activity (Csikzentmikhalyi, M, 2002). This research aims to explore the relationship between NPCs and the state of flow in gaming, understanding how NPCs can be designed to induce and enhance flow experiences for players. By understanding the impact of NPCs on flow, game developers can create captivating and immersive game experiences that maximize player engagement. The development and use of NPCs have evolved significantly over the years, from simple text-based interactions to complex, lifelike characters with their own personalities, goals, and backstories. Today, the application of NPCs is more advanced than ever in that developers can create human-like interactions with the intent of enabling the player into a Flow state where they are immersed in the world. By understanding the impact of NPCs on flow, game developers can create captivating and immersive game experiences that maximize player engagement, leading to a sense of time speeding up during this period of flow, as Marc Wittmann explains (Wittmann, M, 2022). The concept of Flow first originated in the 1990s by psychologist Mihaly Csikszentmihalyi and has gained significant attention in the video game industry due to its potential to enhance the player experience (Oppland, 2016). As defined by Jeanne Nakamura, “Being “in Flow” is the way that some interviewees described the subjective experience of engaging just-manageable challenges by tackling a series of goals, continuously processing feedback about progress, and adjusting action based on this feedback (Nakamura & Csikszentmihalyi, 2002). Flow is an achievable state where your mind is prepared and engaged in the current task at hand and applies to every aspect of your life (Nakamura & Csikszentmihalyi, 2002). As this pertains to game design, NPC’s can be designed using human psychology to incite Flow, giving the player a more memorable and engaging experience whether it is a single-player game designed for immersion or a player vs. player game for balance and teamwork. Non-player characters play a crucial role in creating a sense of immersion and Flow for players in video games by providing a more realistic and dynamic game world. Examples of this can be seen in the popular game Skyrim where NPC’s are coded in a way to give the player user choice in their decisions (Bethesda, 2011). This functionality of user choice adds a more “human-like” and open-ended interaction to the players where they can make choices that affect the state of the world they're immersed in. Taking this a step further, the NPCs also have a day-to-day schedule that they follow giving them a more realistic engagement to the player (Bethesda, 2011). These aspects of NPC behavior significantly contribute to the liveliness and dynamism of the game world, enabling players to perceive the direct consequences of their choices and observe the subsequent impact on both the NPCs and the overall game environment. As Harbing Lou suggests, future NPC designs will prioritize creating an engaging experience rather than focusing solely on their ability to defeat human opponents (Harbing, L, 2017). By adopting such an approach, NPCs not only enhance the player's sense of immersion and flow but also cultivate a more captivating gameplay experience that motivates players to persistently explore and interact with the intricately crafted virtual world. The impact of AI and machine learning on the creation of NPCs in video games cannot be overstated. Game developers have been influenced by these technologies, allowing them to reimagine the design process. A study conducted at Chalmers University of Technology delved into the realm of realistic NPCs, exploring different approaches and their effects on player interaction. (Pihlgren, et al,. 2016) The findings revealed that implementing more complex NPC systems, particularly with advanced pathfinding algorithms, fosters more immersive and meaningful experiences for players (Pihlgren, et al,. 2016). More complex systems can be designed to enhance Flow, notably, the popular title “Middle Earth: Shadows of Mordor” utilizes the Nemesis system to dictate player and NPC interactions

(Monolith Productions, 2014). This system provides the player with a multitude of interactions without the player's input, producing a more natural interaction between the NPC and the actual player. An example of this can observe in the turn-based mechanic of the game where NPC’s dictate the state of the world by overthrowing other NPCs in a carefully calculated system that lets the player know that the world is alive and functioning without direct involvement from the player (Monolith Productions, 2014). In order to further explore the impact of NPCs on the state of flow in gaming and provide practical insights for game developers, this research also includes a study conducted as part of my own capstone project. The study dives into the specific design and implementation of NPCs in a Crisis of DreamCell, prominently following Arzate Cruz & Ramireez Uresti’s understanding of AI as it relates to communicating with the player (Arzate Cruz & Ramireez Uresti, 2017). Aiming to examine how their behaviors and interactions can influence player engagement and immersion, ultimately contributing to the attainment of a state of flow. In the context of the research conducted on the game Crisis of Dreamcell, an examination was carried out to investigate the impact of NPCs on the state of flow. The study involved playtests where participants engaged with the game and interacted with specific NPCs that had varying levels of communication from text to speech. Through careful observation and analysis of participant experiences, the study aimed to explore how NPCs influenced player immersion and the attainment of a flow state. The findings shed light on the significant role of NPCs in enhancing player engagement and emotional involvement, contributing to a heightened sense of immersion and flow within the game. when characters had the appearance of human-like features, players were kore subscribe to speech communications as it was a more human experience rather than text, providing the player with a more immersive experience. These insights provide valuable implications for game developers seeking to create captivating and immersive experiences for players. Several popular video games have successfully implemented NPCs in unique and innovative ways, resulting in enhanced player experience and higher levels of engagement. 2015’s Fallout 4 (Bethesda, 2016) uses a unique system for companion NPCs which gives feedback based on the choices a player makes. For example, when with a less confrontational personality NPC, they are more likely to disagree with your choice to run in “guns blazing” when a more passive approach was also presented. This functionality gives feedback to the player and adds a layer of immersion when the companion the player has chosen does not agree with their actions (Bethesda, 2016). Another example of successful NPC implementation can be seen in the game “The Legend of Zelda: Breath of the Wild” (Nintendo, 2017), where NPCs are designed to have their own schedules and behaviors based on the in-game time and weather conditions. This makes the game world feel dynamic, as players can notice the NPCs going about their daily routines (Nintendo, 2017). Some NPCs might only appear during specific times of the day or in certain weather conditions, This provides players with a sense of discovery and exploration as they uncover new interactions and secrets in the game world (Nintendo, 2017). These innovative uses of NPCs demonstrate the potential for NPC-assisted Flow in games and provide the groundwork for improved iterations of complex systems pertaining to Flow (Pihlgren, et al., 2016; Nintendo, 2017; Bethesda, 2016). Despite their benefits, NPCs pose several challenges for game developers, including finding the right balance between immersion and engagement, as well as avoiding behaviors that may break player immersion. Christopher Madan (2020) explains that the use of tool-assisted human performance benchmarks, such as tool-assisted speed runs, could provide a valuable approach to addressing these challenges and evaluating NPC behavior in video games (Madan, C., 2020). Ming Cao (2020) highlights the potential synergy between game theory and control theory in the realm of intelligent decision-making processes. Additionally, Ming (2020) emphasizes the importance of considering cognitive characteristics and human-in-the-loop control systems in complex cyber-physical-social systems, as exemplified by the impact of self-interested drivers on traffic systems (Cao, M., 2020). The steps to making effective NPCs are numerous and only ever really blocked by time and resources. In a perfect world, developers could make endless possibilities of meaningful interactions between game characters and actual players resulting in more human-like behavior and increased immersion, yet time and resources must be a factor in this equation to balance out interactions that induce Flow and game logic that keep the product still a game. When it comes to developing realistic behaviors in video games, it is essential to strike a delicate balance. Poorly optimized NPC interactions can not only lead to game-breaking bugs but also detract from the overall immersion and Flow of the game (Saagie, 2020.) To avoid such issues, game developers must carefully navigate the challenge of creating engaging and immersive NPCs while also ensuring that they do not negatively impact the player's experience. In this regard, understanding different approaches and implementing effective solutions becomes crucial. For instance, the use of AI and machine

learning has significantly influenced NPC development, allowing for more complex systems and meaningful player interactions (Saagie, 2020.). The behavior of NPCs in video games can significantly impact player emotions and immersion, with NPCs that are Interactive and adaptive often providing a more immersive and engaging experience. In an article by Michael S. Harré (2022), he explores the concepts of introspection and Theory of Mind, explaining how a more precise understanding of psychological ideas can lead to an understanding of strategic interactions (Harré, M. S., 2022). Drawing insights from this article, we can apply these concepts to the realm of video games, specifically by the success of the AI AlphaGo in playing the game of Go (Harré, M. S., 2022). Despite the vast number of possible moves in the game, players, both human and AI rely on a limited number of initial positions to explore their future moves (Harré, M. S., 2022). This approach, rooted in cognitive limitations, is common among game experts, emphasizing the quality of moves considered rather than the quantity (Harré, M. S., 2022). This has been observed in various studies, indicating that players are more likely to feel emotionally invested in the game world when they feel that their actions and decisions are having an impact on the NPCs and the game world itself, creating a more dynamic and responsive gaming experience (Arzate Cruz & Ramireez Uresti, 2017). Video games are supposed to make you feel a certain way when playing them. NPCs that are adaptive to the players' interaction have a greater effect on the players' emotions when playing. The players' emotional state should be considered when designing NPCs to induce Flow. The future of NPC design and implementation will likely involve advanced technologies such as AI and a more opened ended approach to the game world (Sam Cawley, 2023). Skyrim modders are able to enhance the functionality of the NPCs with ChatGPT where the player is able to input their own dialogue and get a completely unique conversation with Skyrim NPCs (Sam Cawley, 2023). Taking this one step further, companies could design their own AI using logic found created in the game world to handle NPC interactions giving players even more opportunities for immersive and personalized experiences. As such, the future of NPC design will likely require a careful balance between innovation and responsibility, ultimately leading to more engaging video game experiences for players. The implementation of NPCs in video games can significantly influence player emotions and enhance player immersion, contributing to the attainment of a state of flow (Wittmann, M, 2022). Game developers can benefit from implementing NPCs in a thoughtful manner, with a focus on enhancing player experience while avoiding potential pitfalls. Emotion and attention are two factors in achieving the Flow state and Developers can use NPCs to stimulate players' emotions and maintain their attention throughout the gameplay experience (Nakamura & Csikszentmihalyi, 2002). In conclusion, the use of NPCs in video games can have a significant impact on player experience, influencing emotions, enhancing immersion, and facilitating the attainment of the flow state. Game developers have the opportunity to leverage NPCs as powerful tools to stimulate player emotions and maintain their attention, thereby creating a more engaging and immersive gameplay experience. By using NPCs thoughtfully and responsibly, developers can maximize the benefits of NPCs by adding cognitive and human-like interactions.

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